Jiahao Cao

jiahao.cao@uth.tmc.edu | Personal Website

Academic Appointments

The University of Texas Health Science Center at Houston Oct. 2024–Present
Postdoctoral Research Fellow *Houston, USA*

Bayesian methodology development, grant writing, and interdisciplinary collaboration.

Education

Renmin University of China Sept. 2019–June 2024
Ph.D. in Statistics *Beijing, China*

Chinese University of Hong Kong, Shenzhen Sept. 2022–Feb. 2023
Visiting Ph.D. Student *Guangdong, China*

University of California, Berkeley Jan. 2019–May 2019
Visiting Student *California, USA*

Nankai University Sept. 2015–June 2019
B.S. in Statistics *Tianjin, China*

Research Identity

I develop statistical methods that combine modeling flexibility with scientific interpretability, with an emphasis on Bayesian inference for complex structured data. My applications and collaborations span public health, environmental science, and sports analytics, where data often exhibit spatio-temporal dependence, high dimensionality, structural constraints, or incomplete observation.

Publications and Manuscripts

* Selected work for which I led or co-led the research.

Published journal articles

- * **Jiahao Cao**, Shiyuan He, Bohai Zhang (2023). Spatial Linear Regression with Covariate Measurement Errors: Inference and Scalable Computation in a Functional Modeling Approach [Link]. *Journal of Computational and Graphical Statistics*.
- * **Jiahao Cao**, Kehe Zhang, Guanyu Hu, Kelly Nelson, Cici Bauer (2026). Bayesian Spatio-temporal Small Area Modeling: A Case Study Investigating the Late-Stage Melanoma Incidence in Texas [Link]. *Journal of the Royal Statistical Society: Series C*.
- Jack Cordes, Cici Bauer, Dana Bernson, Marc R Larochelle, Wenjun Li, **Jiahao Cao**, Olaf Dammann, Thomas J Stopka (2026). The All-Payer Claims Database After the Gobeille Supreme Court Decision: Assessing the Impact on Fatal Opioid-Related Overdose Research [Link]. *Health Services Research*.

Accepted book chapter

- Cici Bauer, **Jiahao Cao**, Kehe Zhang (2026). Bayesian Small Area Estimation for Spatiotemporal Health Data: Methods and Applications. Accepted for publication in *Advances in Public Health Statistics and Data Analytics*, Springer Nature.

Manuscripts under review or revision

- * **Jiahao Cao**, Shiyuan He, Bohai Zhang (2026). Functional BART with Shape Priors: A Bayesian Tree Approach to Constrained Functional Regression [arXiv]. Invited revision submitted to *Bayesian Analysis*.
- * **Jiahao Cao**, Kehe Zhang, Cici Bauer (2026). Bayesian ACCESS for Understanding Latent Epidemic Trajectories from Publicly Released Suppressed Data: Application to U.S. Opioid-Related Overdose Mortality [arXiv]. Submitted to *Journal of the American Statistical Association*.
- Hongyu Chen, **Jiahao Cao**, Bohai Zhang (2026). Spatial Varying Coefficient Model with Covariate Measurement Errors: Bias-Corrected Inference and Efficient Computation. Submitted to *Journal of Computational and Graphical Statistics*.
- **Jiahao Cao**, Qingpo Cai, Lance A. Waller, DeMarc A. Hickson, Guanyu Hu, Jian Kang (2026). What Influences the Field Goal Attempts of Professional Players? Analysis of Basketball Shot Charts via Log Gaussian Cox Processes with Spatially Varying Coefficients [arXiv]. Submitted to *Statistical Analysis and Data Mining: An ASA Data Science Journal*.

Preprints and manuscripts being prepared for submission

- * **Jiahao Cao**, Hou-Cheng Yang, Guanyu Hu (2026). How do professional players select their shot locations? An analysis of Field Goal Attempts via Bayesian Additive Regression Trees [arXiv]. Following peer review at *Journal of the American Statistical Association*, manuscript in revision for submission to *Journal of Computational and Graphical Statistics*.
- **Jiahao Cao**, Alexander Testa, Nicholas Reger, Qian Xiao, Cici Bauer (2026). Nighttime Illumination and Crime Dynamics in Houston: A Longitudinal Cross-Lagged Analysis. Manuscript in preparation for submission to the *American Journal of Public Health*.

Research Interests

Methodological interests

- **Bayesian and nonparametric modeling:** Bayesian additive regression trees and Dirichlet process mixtures.
- **Spatial and spatio-temporal statistics:** Gaussian processes, spatio-temporal point processes, small-area estimation, and network-based spatial models.
- **Structured data modeling and latent-structure discovery:** Compositional data, functional data, high-dimensional regression, and constrained inference; clustering, community detection, and change-point detection.
- **Inference with incomplete or privacy-protected data:** Measurement error, missing data, data suppression, and differential privacy.

Application areas: Public health, disease surveillance, environmental health, public safety, and sports analytics.

Awards and Honors

Outstanding Graduate 2024
Renmin University of China

First Place, Parameter Estimation Sub-competition 2021
KAUST Competition on Spatial Statistics for Large Datasets

First-Class Academic Scholarship 2019, 2020, 2022
Renmin University of China

First Prize, Tianjin 2018
China Undergraduate Mathematical Contest in Modeling

"Gong Neng" Scholarship 2016, 2017
Nankai University

Academic Service

Journal referee: *Journal of Computational and Graphical Statistics*; *Biometrics*; *Annals of Applied Statistics*; *Journal of Biopharmaceutical Statistics*; *Applied Stochastic Models in Business and Industry*.

Teaching Experience

The University of Texas Health Science Center at Houston Spring 2027
Scheduled Teaching Contributions *Houston, USA*

- *Spatial-Temporal Analysis for Population Health Data*: Scheduled to assist with course-material preparation and deliver guest lectures on *Bayesian Count Data Modeling in Public Health*.

Institute of Statistics and Big Data, Renmin University of China
Teaching Assistant *Beijing, China*

- *Computer Skills for Data Science* (Fall 2020, Fall 2022, Fall 2023): Grading and weekly recitation sections.
- *Data Science and Artificial Intelligence Algorithms* (Fall 2021): Grading and weekly recitation sections.
- *Bayesian Modeling and Inference* (Spring 2021): Grading and weekly recitation sections.

Conference Presentations

APHA Annual Meeting & Expo 2026

Oral presentation: *Nighttime Lighting and Crime Dynamics in Houston: A Longitudinal Cross-Lagged Analysis*

Joint Statistical Meetings 2026

Contributed talk: *Bayesian ACCESS for Understanding Latent Epidemic Trajectories*

ICSA Applied Statistics Symposium 2025

Invited speaker: *Functional Bayesian Additive Regression Trees with Shape Constraints: Theory and Application*

EAC-ISBA Conference 2024

Invited speaker: *Functional Bayesian Additive Regression Trees with Shape Constraints*

Workshop on Spatial Statistics, Renmin University of China 2024

Speaker: *Inference and Scalable Computation for Spatial Errors-in-Variables Models*

Joint Conference on Statistics and Data Science in China (JCSDS) 2023

Contributed speaker: *Inference and Scalable Computation for Spatial Errors-in-Variables Models*